PROJECT PORTFOLIO

A planner who turns customer and market insight into results

Service Planning Portfolio

Minkyeong Kim

※ Some figures and names have been altered for confidentiality.

CONTENTS

Contents

01 **Enterprise AI Infrastructure** — AI agent connection hub / MD business ontology / PaaS operations

02 **AI Talent Management** — AI agent training / Fashion AI platform

03 **Internal Business Systems Planning** — Image Platform / RAG search

04 **Strategy Projects (Internal Consulting)** — Naver channel strategy / Padded-jacket rebranding / China brand churn analysis

SUMMARY

Summary

Started in customer and market analysis for fashion brands; now building the platform that connects company data to AI agents and designing the company-wide AI transformation. From the numbers to the shipped service, end to end.

No.	Part	Project	Period	Key result
01	Enterprise AI Infrastructure	**Knowledge Graph-based Enterprise AI Agent Connection Hub**	2025.11 – ongoing	**160** Agents (business web apps) completed
02	Enterprise AI Infrastructure	**Building the Ontology: Adding a Business-Concept Layer to the Data Catalog**	2026.11 – ongoing	**markdown rate · reorder quantity · reorder timing · allocation · main sales channel** Wider answer scope (judgment questions)
03	Enterprise AI Infrastructure	**PaaS-based Service Operations**	2026.03 – ongoing	**311** Deployments
04	AI Talent Management	**Company-wide AI Agent Training**	2026.02 – ongoing	**500 in total** AI agent training attendees
05	AI Talent Management	**Fashion AI Platform: Web-App Sharing, Recommendation & Submission Portal**	2026.08 – in operation	**160 total** Agents (web apps) uploaded
06	Business Systems	**Company-wide Integrated Image Platform**	2025.09 – 2026.03	**6 days → ≤1 hour** Listing lead time
07	Business Systems	**Internal Strategy-Document RAG Search**	2025.01 – 2025.07	**98%** Budget saved
08	Strategy Projects	**Redefining the Naver Channel Sales Strategy**	2024.01 – 2024.02	**137%** Live-commerce conversion growth
09	Strategy Projects	**Sports Brand Padded-Jacket Rebranding Strategy**	2024.11 – 2024.12	**100%** Sales growth among customers under 30
10	Strategy Projects	**Customer Churn Analysis for a Premium Brand in China**	2025.02 – 2025.04	**+8%** Sales growth vs. same month last year

Part 01 | Enterprise AI Infrastructure

01

Knowledge Graph-based Enterprise AI Agent Connection Hub

A knowledge-graph-based enterprise AI agent connection hub where a plain-language request is enough to build a web app connected straight to the company DW

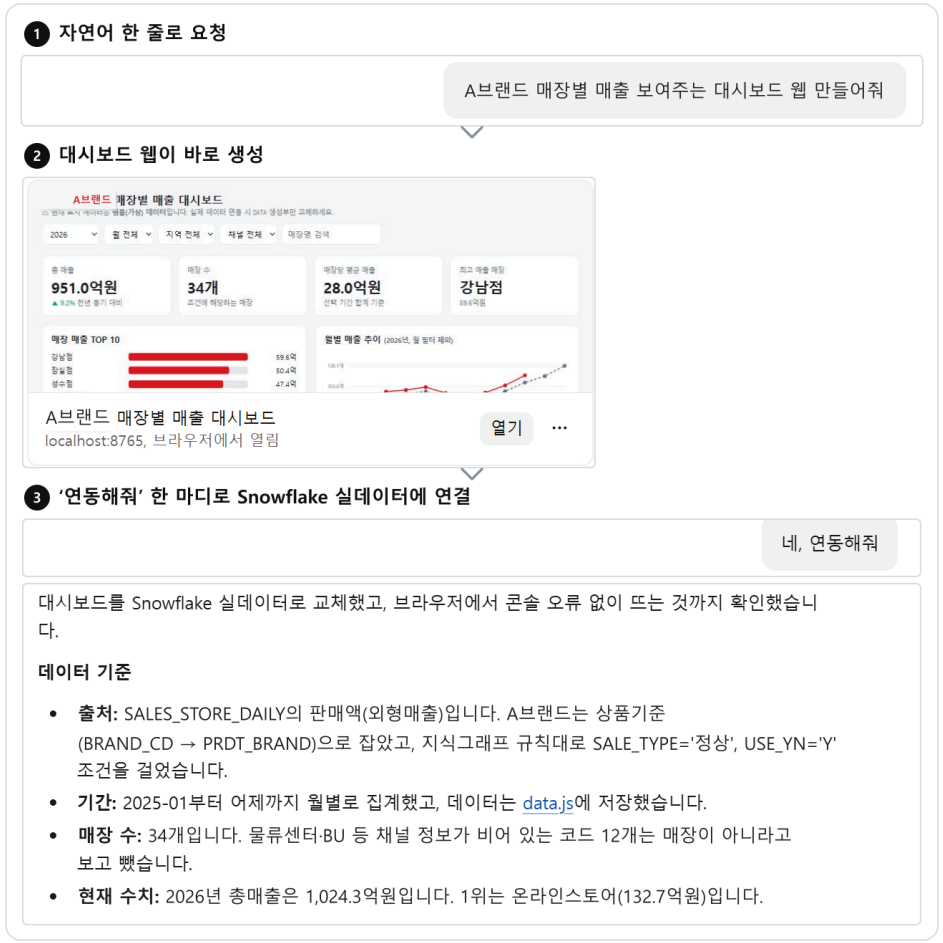
Period 2025.11 – ongoing

Knowledge graph | MCP | Text to SQL

Knowledge Graph-based Enterprise AI Agent Connection Hub

A knowledge-graph-based enterprise AI agent connection hub where a plain-language request is enough to build a web app connected straight to the company DW

Building a dashboard by vibe coding with this tool



Vibe-coding demo: one word, 'connect it', links it to live Snowflake data

Period

2025.11 – ongoing

Area

Knowledge graph · MCP · Text to SQL

Stack & tools

MCP · Claude Code Skill · Snowflake · Neo4j · Cypher queries

Purpose

1. Build web apps connected to the company DW with plain-language requests alone

2. Query BI data in plain language

Role & contribution

1. Built a mapping dictionary between business terms (brand, season, store) and DW tables and columns: organized synonyms and reference values so the same question gets the same answer
2. Defined requirements for the MCP server linking the graph and the DW: agreed the tool list, input/output formats and read-only permission scope with developers
3. Designed the plugin rollout process: wrote the install guide, connector setup steps and a troubleshooting guide for common errors

Results

Agents (business web apps) completed

160

as of Sept 2026 (first month after launch)

Work hours saved by agents

8,180 h / month

as of Sept 2026 (first month after launch)

Time to query in-house (BI) data

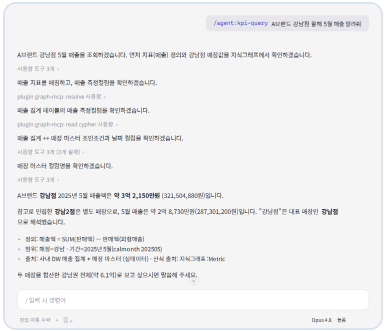
85% less

AS-IS 27 min → TO-BE 4 min

Knowledge Graph-based Enterprise AI Agent Connection Hub

1 Problem	2 What I did	3 Result
Web apps built by employees could not connect to the company DW, so they saved little real time, and a single BI data query took 27 minutes on average; action was needed.	**Put metric definitions in a knowledge graph so a plain request builds a DW-linked web app.**	**160 agents built and 8,180 work hours saved per month (Sept 2026).**

Querying BI data in plain language



The kpi-query skill: natural-language question → metric definition resolved from the knowledge graph → DW query → answer with sources

Background

1. Web apps built by employees could not connect to the company DW, so they saved little real time; the aim was to connect individually built web apps to in-house data while keeping security issues to a minimum.
2. Querying in-house BI data took 27 minutes on average; the aim was to cut that time by 75% with AI.

System architecture

This hub puts agents on top of a knowledge graph. Built on a Neo4j knowledge graph, it processes each question in the order organization instructions → hooks → skills → MCP tools → gateway → knowledge graph and internal data.

1. Client Layer
The company plugin in Claude Desktop, Claude Code or Cowork

- **Claude Desktop · Code · Cowork**
- **Company plugin** — bundle of skills, hooks and MCP connectors
- **Org connector registration** — deployed per organization by admins
- **SSO login** — queries run with the user's own permissions

2. Policy & Context Layer
Org instructions and hooks read the context graph

- **Org instructions** — rules for everyone, e.g. what to do when an MCP connection fails
- **Session-start hook** — personalization, memory and in-house PaaS development rules
- **Pre-question hook** — adds past corrections, golden SQL and company decisions from the context graph
- **Safety principles** — read-only · own permissions · no personal-data columns · verify the person in charge is still employed

3. Agent Runtime
The Claude model picks a skill and calls tools in order

- **Claude model** — reasoning · SQL writing · answer writing
- **Skill triggers** — skills selected by keyword and request type
- **Memory** — per-user preferences and team rules kept as files; new definitions are recommended for knowledge graph registration

4. Capability Layer (skills)
Skills hold no rules; they only set how to query the graph and which tools to call

- **Natural-language KPI skill** — one graph lookup → SQL → read-only DW run
- **Weekly report search skill** — finds brand weekly reports via the graph's vector index
- **Internal admin guide skill** — expenses, policies, general affairs and other internal knowledge search
- **App build & deploy skill** — building and preparing apps for deployment on the company standard

Knowledge Graph-based Enterprise AI Agent Connection Hub

5. Tool Binding Layer (MCP tools) The tools skills actually call

Knowledge graph lookup tool	Company DW query tool	HR (attendance) tool	Mail & calendar tool
metric meaning · graph reads · weekly report and admin search	runs Snowflake queries (read-only)	leave info, requests, approval lines, approve, cancel	read, send and reply to mail; create and update events

6. Knowledge Graph & Data Layer

Through the in-house MCP gateway, the DW, M365 and HR are read by the knowledge graph's definitions

Neo4j knowledge graph	Snowflake DW	M365 Graph	In-house PaaS
metric definitions, formulas, joins, golden SQL	standard data models (aggregate tables)	mail and calendar	one-click deployment of employee-built apps

How a question is processed

1 Question
Ask in everyday words, e.g. “What were Brand A Gangnam's sales this May?”

2 Knowledge graph lookup
One graph lookup returns the metric definition and joins; SQL is written only from them

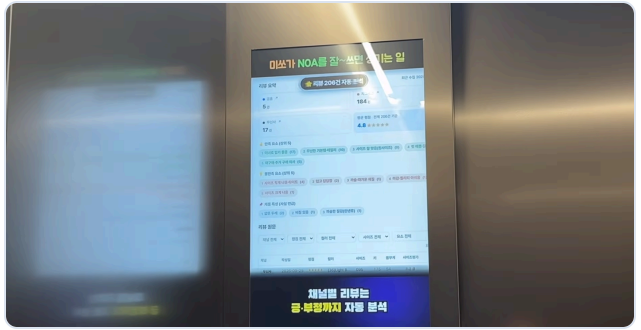
3 Answer with sources
Read-only DW query with the user's own permissions; definition, scope and sources attached

4 Learning from corrections
A correction like “No, from 25 January” is saved to the context graph and applied from the next question

Process

01 Knowledge graph design and build › 02 MCP servers connecting the graph and the DW › 03 Skills and hooks

04 Plugin rollout and offline training › 05 Weekly use-case broadcasts across affiliates



Step 5 in practice: a weekly use-case video broadcast on office elevator screens (automatic positive/negative review analysis by channel)

Part 01 | Enterprise AI Infrastructure

02

Building the Ontology: Adding a Business-Concept Layer to the Data Catalog

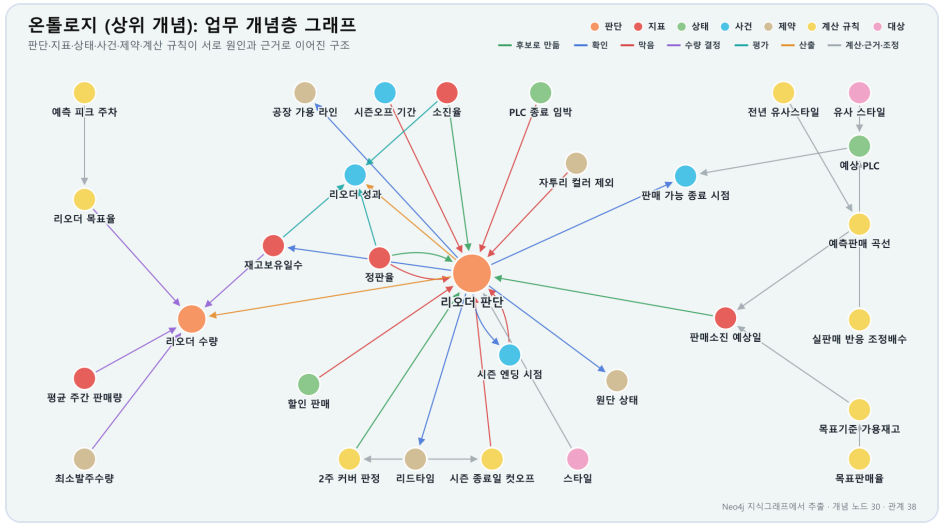
From a semantic layer centered on DW tables and columns to a business ontology with business-judgment concepts added

Period 2026.11 – ongoing

Ontology | MD knowledge | Knowledge graph

Building the Ontology: Adding a Business-Concept Layer to the Data Catalog

From a semantic layer centered on DW tables and columns to a business ontology with business-judgment concepts added



Ontology (upper concepts): the business-concept layer where decisions, metrics, states, events, restrictions and calculation rules are linked as causes and grounds (30 nodes · 38 relations)

Period

2026.11 – ongoing

Area

Ontology · MD knowledge · Knowledge graph

Stack & tools

Neo4j · Cypher queries · Snowflake

Purpose

Extends the range of answers to judgment questions by building an ontology that holds business know-how on top of the existing data catalog (semantic layer)

Role & contribution

1. Organized business concepts and causal relations through interviews with practitioners
2. Defined the concept-table and relation-table formats (direction, attributes, exceptions)

Results

Wider answer scope (judgment questions)

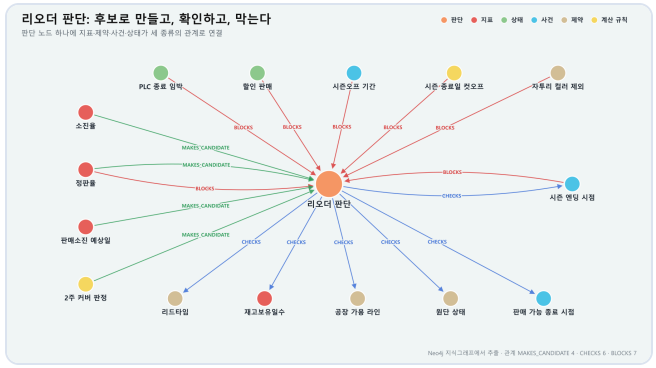
markdown rate · reorder quantity · reorder timing · allocation · main sales channel

Building the Ontology: Adding a Business-Concept Layer to the Data Catalog

1 Problem	2 What I did	3 Result
Company judgment rules weren't in the data catalog, so users re-entered them each session.	**Drew out five weekly MD decisions in interviews and designed an ontology layer.**	**AI now answers judgment questions such as markdown rate and reorder quantity and timing.**

Glossary

Full-price sell-through · Share of stock sold at full price, without discounts
Sell-out rate · Share of received stock that has been sold
PLC · Product life cycle: introduction, growth, peak, decline
Reorder · Ordering more of a product that is selling well



Reorder decision: metrics, restrictions, events and states make it a candidate (MAKES_CANDIDATE), are checked (CHECKS) and block it (BLOCKS)

Background

The company's unwritten rules and ways of working were not in it, so even with a semantic layer, users had to enter the company's judgment criteria again in every session.

System architecture

The existing knowledge graph (semantic layer) serves as the DW data-link layer, while MD experience knowledge such as 'fit complaints → size issue → more returns → lower full-price sell-through' is added to support faster decisions

1. Ontology (upper concepts · schema)

Nouns become concepts, verbs become relations. Concepts are grouped into buckets (labels) from the MD world, not the DW.

Metrics
sell-out rate · full-price sell-through · days of stock · early sell-through

States / stages
PLC early · growth · peak · decline

Decisions
reorder · channel focus · markdown timing · next-season orders

Constraints
days of stock · lead time

Timing / events
season end · off-season sale · forecast peak passed

2. Knowledge graph (lower concepts · real data and relations)

Says where each concept is visible in data. Nothing built so far is thrown away; it keeps the upper layer from misreading data.

Full-price sell-through → DW sales metrics

Fit complaints → review text · platform detail ratings

Guarding facts
the repair/inspection CS channel is not customer reviews · company-mall photos are not collected · jewelry accumulates lines per receipt

Building the Ontology: Adding a Business-Concept Layer to the Data Catalog

How a question is processed

1

User judgment query

“Why isn't this style selling? What should we do?”

2

Find the decision node

Check which PLC stage it is in, such as introduction or decline.

3

Seen in data as

Confirm exact numbers through the lower-level knowledge graph: full-price sell-through from sales metrics, fit complaints from review-text tables.

Process

01
Interview SO/MD staff (judgment criteria in the work)

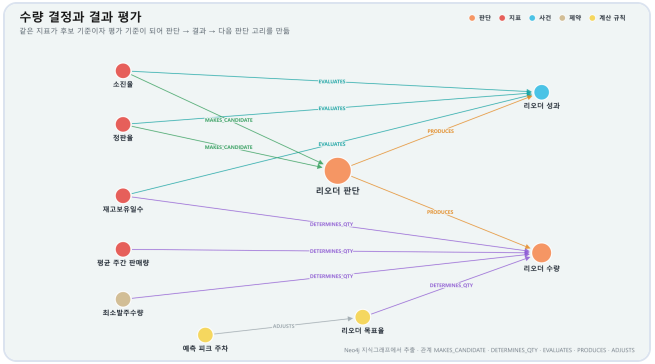
›

02
Write concept and relation tables

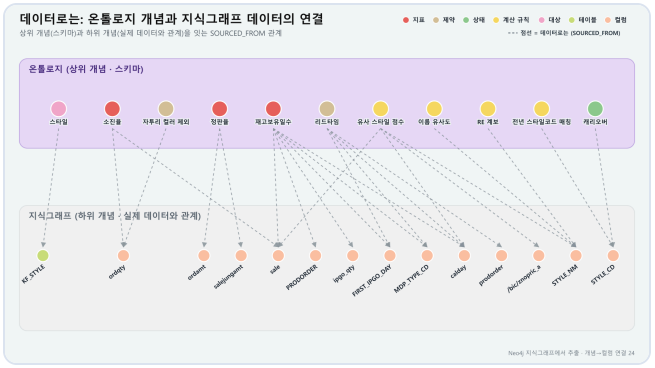
›

03
Migrate to Neo4j

04
Connect the Neo4j MCP



Quantity and evaluation: the same metrics serve as candidate criteria and evaluation criteria, forming a decision → result → next decision loop



Seen in data as: SOURCED_FROM relations tie upper concepts (ontology) to lower concepts (knowledge graph) (24 links)

Part 01 | Enterprise AI Infrastructure

03

PaaS-based Service Operations

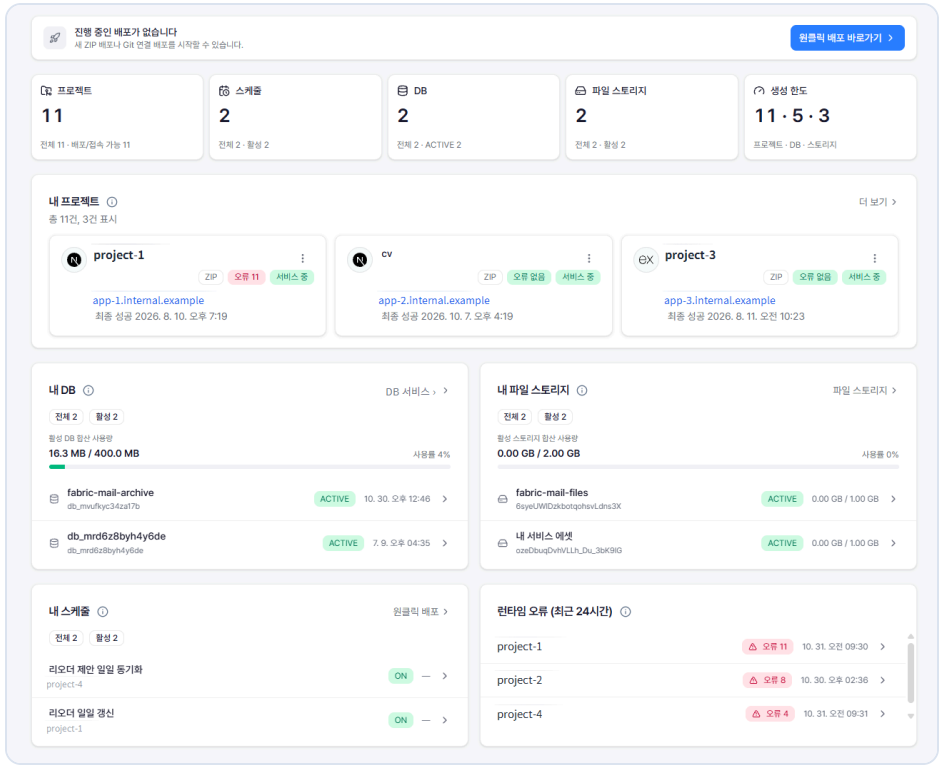
An in-house PaaS that deploys the apps non-developers build and provides the DB, file storage and schedules they need

Period 2026.03 – ongoing

PaaS | One-click deployment | Self-service

PaaS-based Service Operations

An in-house PaaS that deploys the apps non-developers build and provides the DB, file storage and schedules they need



The status of the web apps each person has deployed, as seen on their own dashboard

Period

2026.03 – ongoing

Area

PaaS · One-click deployment · Self-service

Stack & tools

Next.js · Serverless · CloudFront · Claude Code Skill

Purpose

An in-house PaaS where line staff deploy the web tools they build without developer help and get operations support. Overall deployment status, per-app MAU and call counts, and deployment domains are managed in one place.

Role & contribution

1. Wrote, distributed and taught the one-click deployment guardrails (rules for AI coding tools)
2. Designed Claude Code skills (requirements, triggers, behavior) that check a project against PaaS rules before deployment
3. Turned recurring deployment and operations inquiries into skill features

Results

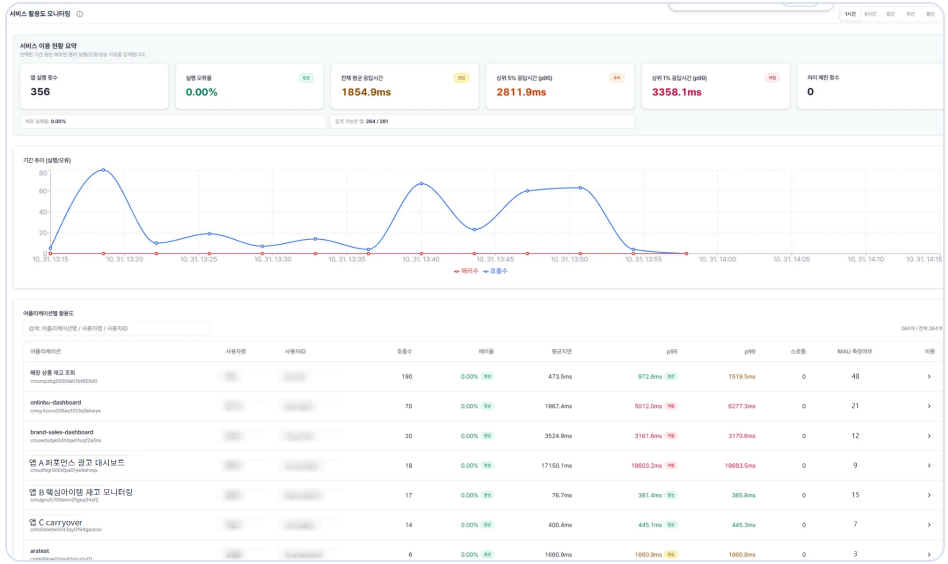
Deployments	Deployment success rate
311	**90%**
	n=235 deployments

PaaS-based Service Operations

1 Problem	2 What I did	3 Result
Employee-built web apps sat on separate servers and outside DBs, hiding data outflow.	**Built an in-house PaaS for deployment, DBs and schedules, plus guardrails for AI tools.**	**311 deployments with a 90% deployment success rate (n=235).**

Glossary

MAU · Monthly active users: people who used an app in a month



Admin view of every deployed web app

Background

Web services built by individual employees were deployed to individual servers, which made company information hard to manage and made it hard to see which company data left through external DBs and external hosting.
The in-house PaaS was built to bring deployment inside the company, manage it in one place, and consolidate internal information, security and cost.

Core features

Core feature 01

One-click deployment

Upload a ZIP or connect a Git repository and the app deploys. Guardrail rules for AI coding tools were shipped first to raise the success rate.

ZIP and Git deployment

Deploy by uploading a finished ZIP or connecting a Git repository. Each project card shows deployments in progress, the last successful time and error counts.

Deployment guardrails

Rules that AI coding tools such as Cursor and Claude Code follow when generating code (platform and stack choices, safe defaults), distributed as a document to cut failed deployments.

PaaS-based Service Operations

Pre-deployment skill

A Claude Code skill checks project structure, environment variables and build settings against PaaS rules before deploying.

Creation limits

Project, DB and storage limits are shown so resources are planned.

Core feature 02

Operations, security and domains on one screen

Usage of each deployed app (MAU and call counts), data-security checks, deployment domains, schedules, DBs and file storage are all managed from one dashboard.

MAU and call counts per app

Monthly active users and API call counts are shown for every deployed app, separating the apps that are really used from the ones to retire and putting operating results in numbers.

Data-security check

Company data must stay in the internal DB, so each app is checked before deployment for connections to any external database, closing the path for company data to leave.

Deployment domain management

Domains are issued, changed and revoked per app, and which domain belongs to which app and owner is managed in one place.

Projects, schedules, DB and file storage

Deployed apps with their service status, jobs that run at set times, and the databases and file stores apps use are created and monitored here.

Part 02 | AI Talent Management

04

Company-wide AI Agent Training

Small, step by step, repeatedly: employees build AI agents hands-on

Period 2026.02 – ongoing

AI agent training | Training | Onboarding

Company-wide AI Agent Training

Small, step by step, repeatedly: employees build AI agents hands-on



AI agent-building training: one-click deployment guide session

Period

2026.02 – ongoing

Area

AI agent training · Training · Onboarding

Purpose

Designed training, onboarding and operational support to reach a 20% AX conversion rate across the organization.

Role & contribution

1. Designed the curriculum and job-specific hands-on exercises
2. Built the training content: level-based concept training and hands-on sessions (building a web dashboard)

Results

AI agent training attendees
500 in total

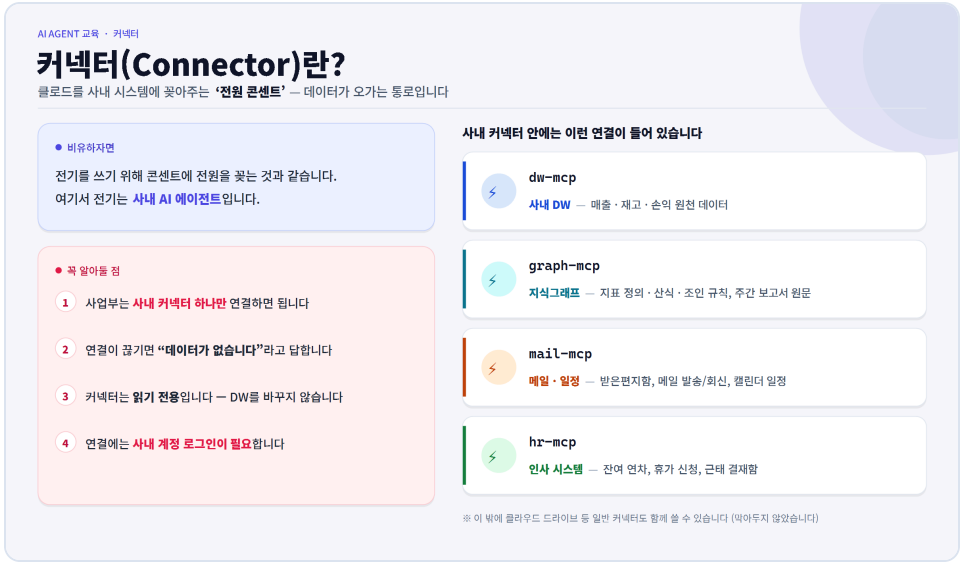
Each attendee finishes one web-based tool of their own by the end of every session
1 per session

Company-wide AI Agent Training

1 Problem	2 What I did	3 Result
The goal was 20% AX adoption, but most employees didn't know where to start.	**Designed small, level-based, repeated classes and a 6-step agent-building practice.**	**500 people trained in total, each finishing one web tool of their own per session.**

Glossary

Hands-on session · A practice class where people build things themselves



Connector training material

Background

To reach 20% AX adoption in-house, small-group, level-based offline training is repeated for employees who do not know where to start.

Process

01 Hands-on agent training › **02** Build & deploy support

Training content

Part 1

Level-based concept training

Because skill levels differ widely, one lecture is too hard for beginners and dull for experienced staff, so classes are split by level and kept small. Concept training is the preparation for the hands-on session that follows.

AI literacy basics

Fourteen slides on AI fundamentals and what matters at work. Security principles and one-click deployment guardrails rewritten in plain company language.

Classes by level

Beginner, intermediate and advanced classes teach concepts at the right level, removing the problem of a single lecture being hard for beginners and dull for experienced staff.

Company-wide AI Agent Training

In-house AI agent 2.0 training decks

Classroom decks in two sessions: session 1 (concepts, 41 slides) and session 2 (hands-on, 30 slides).

Connector concept

Explains connectors with a power-outlet analogy instead of developer terms, covering four things to know and the four connections inside a connector.

Part 2

Hands-on sessions

Right after concept training, even first-time attendees turn a problem from their own work into one agent and publish it. Completion means one web-based tool of their own by the end of every session.

6-step agent-building practice

Understand AI → define the work problem → write a prompt spec → prototype with Claude Code → security check → prepare to share and deploy. Each step has an output and check questions, with first-agent scenarios for MD, sales, design and CS.

5-step vibe-coding guide

Decide what to build → build screens by vibe coding → connect BI data → prepare code for deployment → deploy. A usage-guide prompt makes the builder state operating logic, team tacit rules and KPIs first.

One-click deployment

The finished tool is deployed to the in-house PaaS in one click, with a deployment guide session covering guardrail rules and the deployment procedure.

Build and deploy support after the session

FAQ, 1:1 inquiries, data requests and offline help sign-up resolve whatever blocks people after the session.

Part 02 | AI Talent Management

05

Fashion AI Platform: Web-App Sharing, Recommendation & Submission Portal

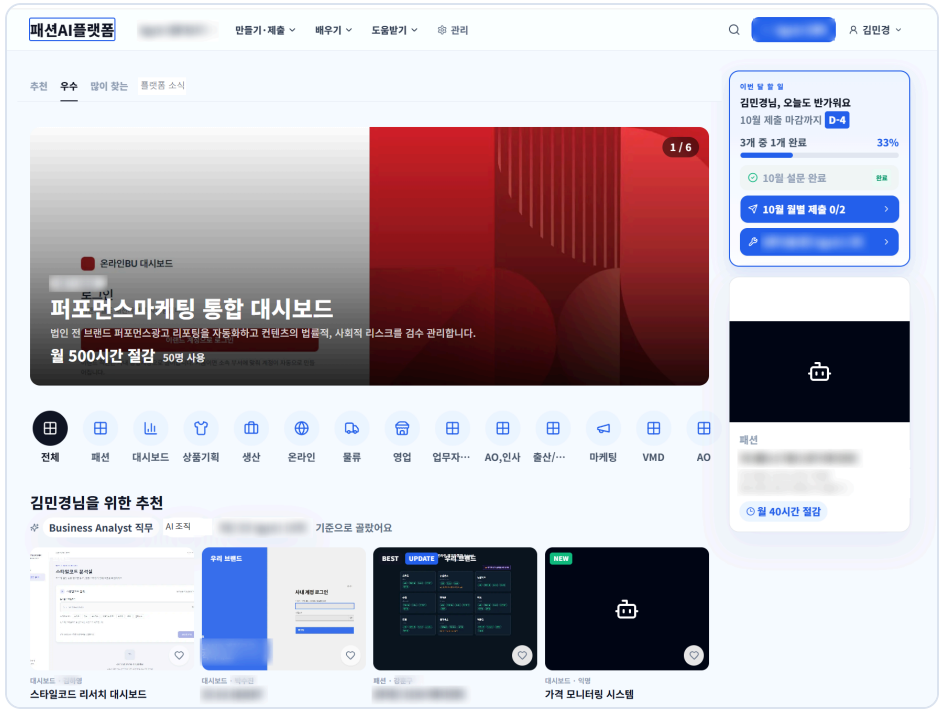
Where employees publish business web apps built with the in-house AI platform, get similar apps recommended by job, and submit results

Period 2026.08 – in operation

Registration & submission | Next.js

Fashion AI Platform: Web-App Sharing, Recommendation & Submission Portal

Where employees publish business web apps built with the in-house AI platform, get similar apps recommended by job, and submit results



Fashion AI Platform home: best web-app banner · categories · monthly to-do sidebar

Period

2026.08 – in operation

Area

Registration & submission · Next.js

Purpose

Employees upload the business web apps they built with the in-house AI platform and browse apps from the same job, and check their own AI capability level

Role & contribution

1. Service planning and information architecture
2. User flow design: UI/UX and menus such as issuing a PAT, so people use the site without friction
3. Monthly submission flow design

Results

Agents (web apps) uploaded

160 total

since 2026.09

Fashion AI Platform: Web-App Sharing, Recommendation & Submission Portal

1 Problem	2 What I did	3 Result
Business web apps built by employees, and their results, were scattered.	**Planned an in-house portal and user flow for web-app registration and monthly results.**	**160 agents (web apps) uploaded in total, with AX data building up automatically.**

Background

Business web apps built with the in-house AI platform multiplied, but who built what and what it achieved was scattered.
Register the web app → submit monthly results → evaluate → train now runs as one flow that records AX status automatically.

Core features

Core feature 01

Home: tasks and recommended reference agents the moment you arrive

The first screen shows this month's tasks together with web apps built by colleagues in the same job that fit me

Best web-app banner

Web apps with the BEST badge rotate in a six-slide banner with monthly hours saved and users

Job and category icon filter

One row of job icons (Fashion, Dashboard, Merchandising, Production…); tap to filter web apps

This month's tasks sidebar

Deadline D-day, progress and survey status pinned on the right of every screen

Core feature 02

Web-app registration and monthly submission

A form that states what changed in numbers, plus monthly results; the source data for evaluation and the talent pool

4-part form

① Description ② Usage guide ③ What changed (hours saved, KPI) ④ Access. Missing fields listed with reasons

Usage-guide prompt: eight items that capture know-how

Paste the prompt into the AI for a 1,500+ character guide; operating logic and work tips matter most

Fashion AI Platform: Web-App Sharing, Recommendation & Submission Portal

Core feature 03

Requests and inquiries that accumulate automatically

Missing data or stuck on usage? Request or ask in the platform; recurring questions become FAQ

Data requests

Request what data is needed and why; tracked by status and turned into DW and graph tasks

FAQ that builds itself

Answers to recurring questions go to the FAQ on home and Help, so repeats fall

Admin morning digest email

Every day at 8 a.m., new accounts, submissions, uploads and survey responses emailed to admins

Part 03 | Internal Business Systems Planning

06

Company-wide Integrated Image Platform

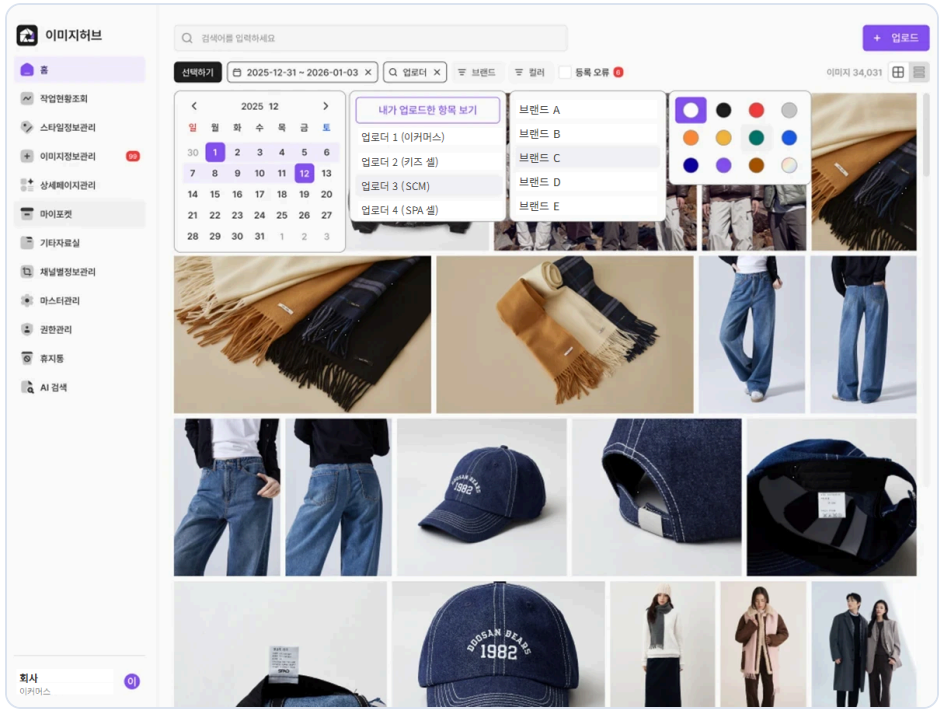
Unified management of every image in the company, with a flow built from image registration to delivery to online channels

Period 2025.09 – 2026.03

Integrated image platform | Product information | AI channel listing

Company-wide Integrated Image Platform

Unified management of every image in the company, with a flow built from image registration to delivery to online channels



Integrated image platform home: upload date, uploader, brand and color filters plus a ‘registration error’ badge locate any of tens of thousands of images in the grid

Period

2025.09 – 2026.03

Area

Integrated image platform · Product information · AI channel listing

Purpose

Manage every product image in the company (model shots, flat-lay shots, thumbnails) in one company-wide integrated image platform, and let AI adapt registered images and product data to each channel's specs and send them straight to online channels. The goal is to end re-entering the same information channel by channel.

Role & contribution

1. Defined what the integrated image platform manages: product code, name, material, color, size, price, inventory and images (model, flat-lay, thumbnail) as the single source of product truth
2. Mapped where data is created and how it flows across SAP, BI, local storage, Google Drive and email
3. Designed the generative-AI workflow: per-item keyword extraction, channel-specific names and promotional copy, image spec adaptation
4. Turned interviews with MD, photo and ops teams into requirements for listing procedures and per-channel rules

Results

Listing lead time

6 days → ≤1 hour

from receiving to on sale

Standard listings automated

100%

only events and promotions remain: 80% of time saved

Company-wide Integrated Image Platform

1 Problem	2 What I did	3 Result
Scattered product data was re-keyed by hand per channel: 54% of repetitive work.	**Designed a flow where a company-wide integrated image platform and AI format listings and send them to every channel.**	**Listing lead time cut from 6 days to at most 1 hour; standard listings 100% automated.**

Glossary

Lead time · Time from receiving stock to going on sale

Style code · A unique ID number given to each product

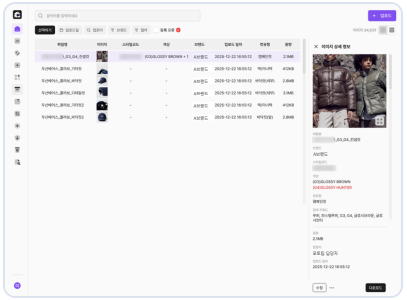


Image registration review and detail check

Background

Interviewing MD and AMD staff for three weeks and surveying their work showed that 54% of simple, repetitive work was ‘product listing and management’ (1,865 hours per month). The two root causes were ① product data scattered across SAP, BI, local folders, drives and email, and ② manual adjustment to each channel's different specs.

System architecture

Listing after receiving runs as one flow: images enter the platform, AI builds detail pages, per-channel agents fit each format and every channel receives the listing.

Appendix 1 Integrated image platform pipeline: product data in → AI processing (keywords, channel listings) → OMS → online channels

1. Bulk upload

Drag-and-drop bulk upload; re-uploads notify the detail-page AI.

- **Drag-and-drop bulk upload** — style code, color and cut type recognized from the file name
- **Re-upload** — added model shots etc. → the page-generation AI notified
- **Error badge** — unmatched files flagged on the home screen

2. Image management

Type, traits and product data turn each image into a searchable asset.

- **① Image type** — front/back/detail, product/model/concept shot
- **② Product traits** — material · mood
- **③ Style-code link** — SAP data (name, price) joined with image platform disclosure data

3. Auto-editing

Images are adjusted automatically to channel specs.

- **① Resizing** — size · ratio
- **② Retouching** — brand logo removal, background removal and more

4. Search

Search by product data, similar image or plain language.

- **Product-information search** — style code · item name · color
- **Similar-image search** — find images with an image
- **AI search** — natural-language search for product images

5. Download

Select the images you need and download them in bulk.

- **Select and bulk download**
- **My Pocket** — keep frequently used images

Company-wide Integrated Image Platform

6. Share Bundle only the images you want to share and edit.

- Shared image bundles
- Edit inside a shared bundle

Product listing flow

1 PLM product registration
Base data registered in PLM flows into the image platform automatically.

2 Receiving and inventory → new-product alert
Receiving triggers an automatic new-product alert to the listing team.

3 Shooting · 3D generation → integrated image platform upload / API
Shots (later 3D renders) go straight to the integrated image platform; no more Google upload or email.

4 Detail page production (AI)
AI generates models, sales copy, detail pages and SEO tags.

5 Thumbnail and detail-page upload
Thumbnails and detail pages attach to the style code automatically.

6 Per-channel data-optimization agent
An agent converts names, categories and image specs for each channel.

7 Listing-hub registration → all channels
One listing goes to every channel: the company mall, Naver, Musinsa and more.

8 Monitoring dashboard
Blocks 'received but never on sale' errors; people only review.

Business process: AS-IS → TO-BE, and the five changes

Appendix 2 [AS-IS] The old listing process: the same information re-typed by people across five teams. The seven red boxes are the bug points

Appendix 3 [TO-BE] The image platform flow after PLM: new-product alerts go out automatically, shots and 3D renders enter through the hub, AI builds the pages, a per-channel agent fits each format and the listing hub sends to every channel. Every red box (bug point) is gone

[AS-IS] The previous business process — seven bug points

1. Merchandising registers the product in SAP
2. Logistics registers receiving and inventory
3. Listing team digs through inventory to find new products — Bug
4. Lists the product without photos
5. Requests samples from logistics
6. Measures and checks inner tags — Bug
7. Makes disclosure images — Bug
8. Photo team shoots
9. Google upload and email — Bug
10. Detail page production — Bug
11. Thumbnails and pages entered
12. Company mall image upload — Bug
13. E-commerce team lists separately in the company mall, Sabangnet and the listing hub — Bug
14. Sent to Cafe24, Naver, Musinsa, Zigzag

Five teams passed the same information along, re-typing it each time.

Company-wide Integrated Image Platform

[TO-BE] After the system — one image platform flow

1 PLM product registration	2 New-product alert as soon as receiving and inventory are registered	3 Shooting / 3D generation
4 Integrated image platform upload / API	5 Detail page production (AI)	6 Automatic thumbnail and detail upload
7 Per-channel data optimization agent	8 Listing hub registration	9 Sent to all channels

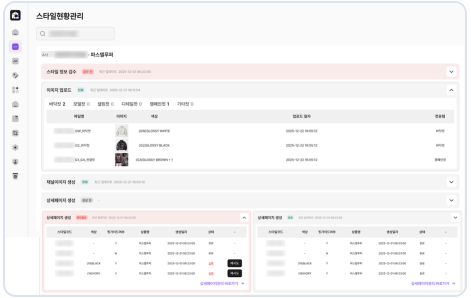
Re-typing steps disappear; the listing team only reviews and handles exceptions.

Change ① Lead time from 6 days to 1 hour	Listing lead time from receiving to on sale: average 6 days → at most 1 hour.
Change ② People no longer intervene	Standard listings 100% automated; people review and give AI-logic feedback. Only events and promotions remain, saving 80% of time.
Change ③ Take the photo and AI does the rest	The page is created and listed as the shot is taken; with 3D modeling, products go on sale at planning without a shoot.
Change ④ No more 'received but not on sale'	A monitoring dashboard completely blocks products received but never put on sale.
Change ⑤ Data as an asset, and what it enables	All sales-side product data feeds AI training, enabling AI merchandising, recommendations and short-form content, and global channel expansion.

Process

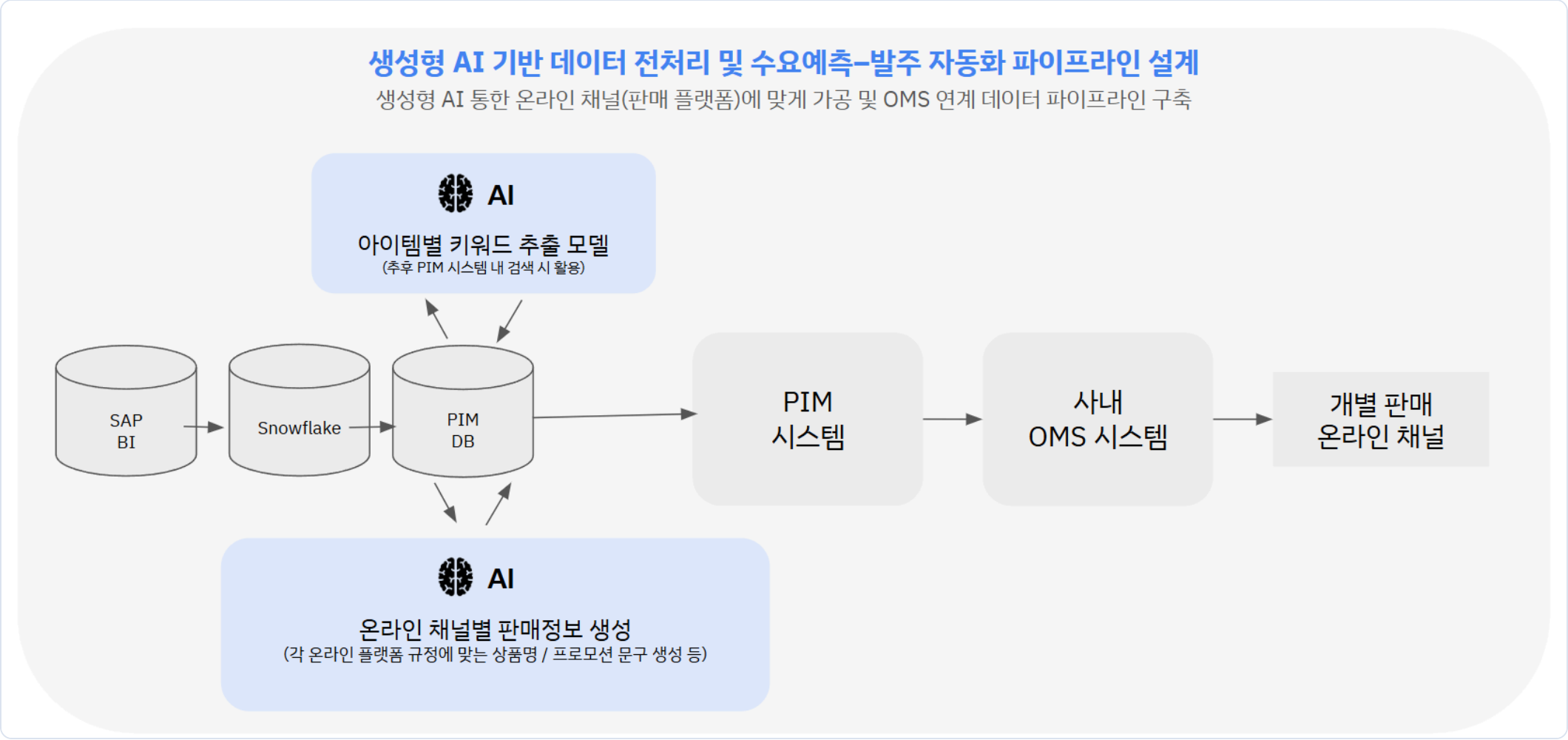
01 3-week MD/AMD work census › **02** Define the two root causes › **03** Image platform attributes and data flow

04 AI detail pages and channel-optimization agent › **05** Listing hub → all channels automatically



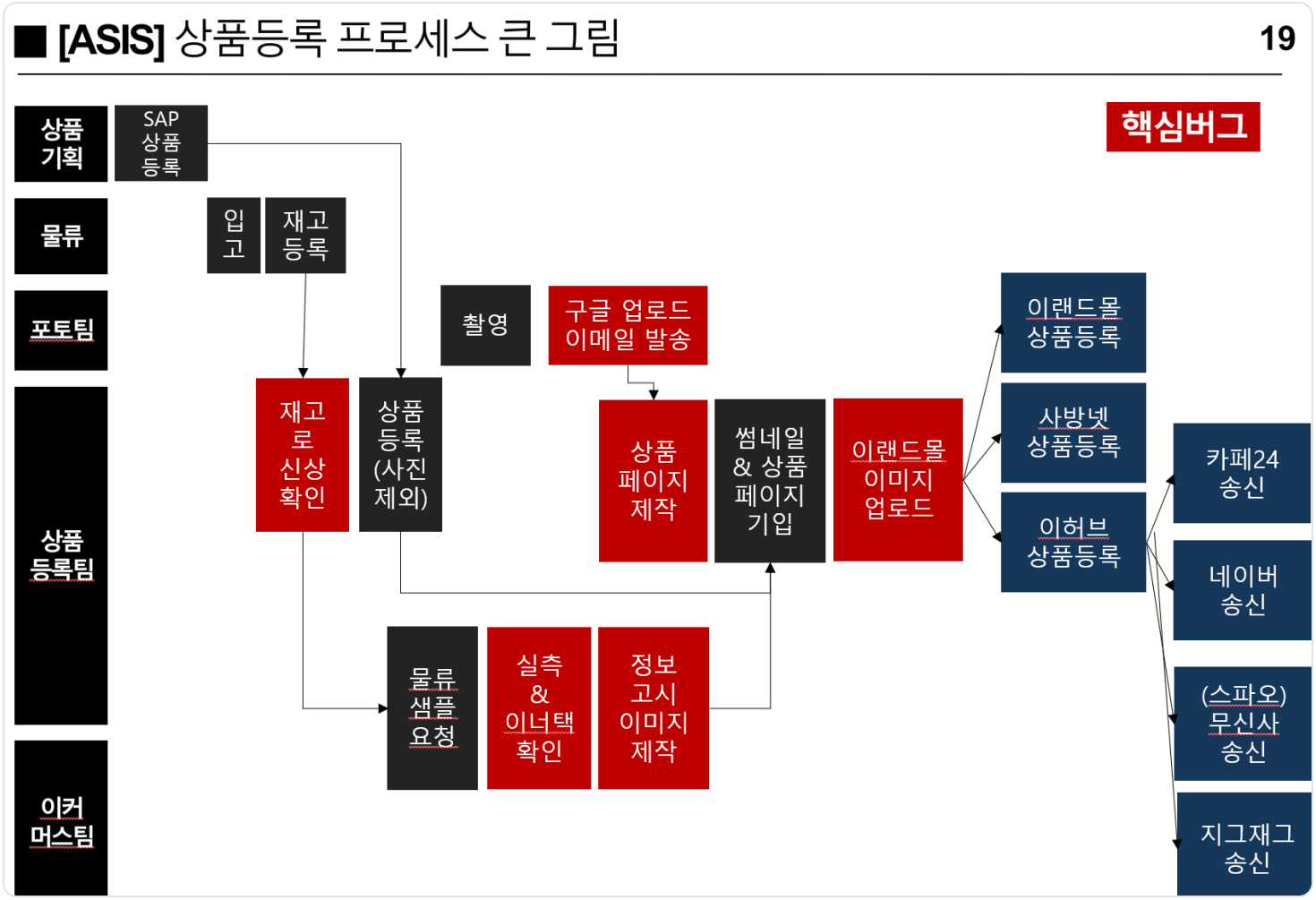
Style status: image upload → automatic per-channel image editing → review of the detail-page generation stage

Company-wide Integrated Image Platform



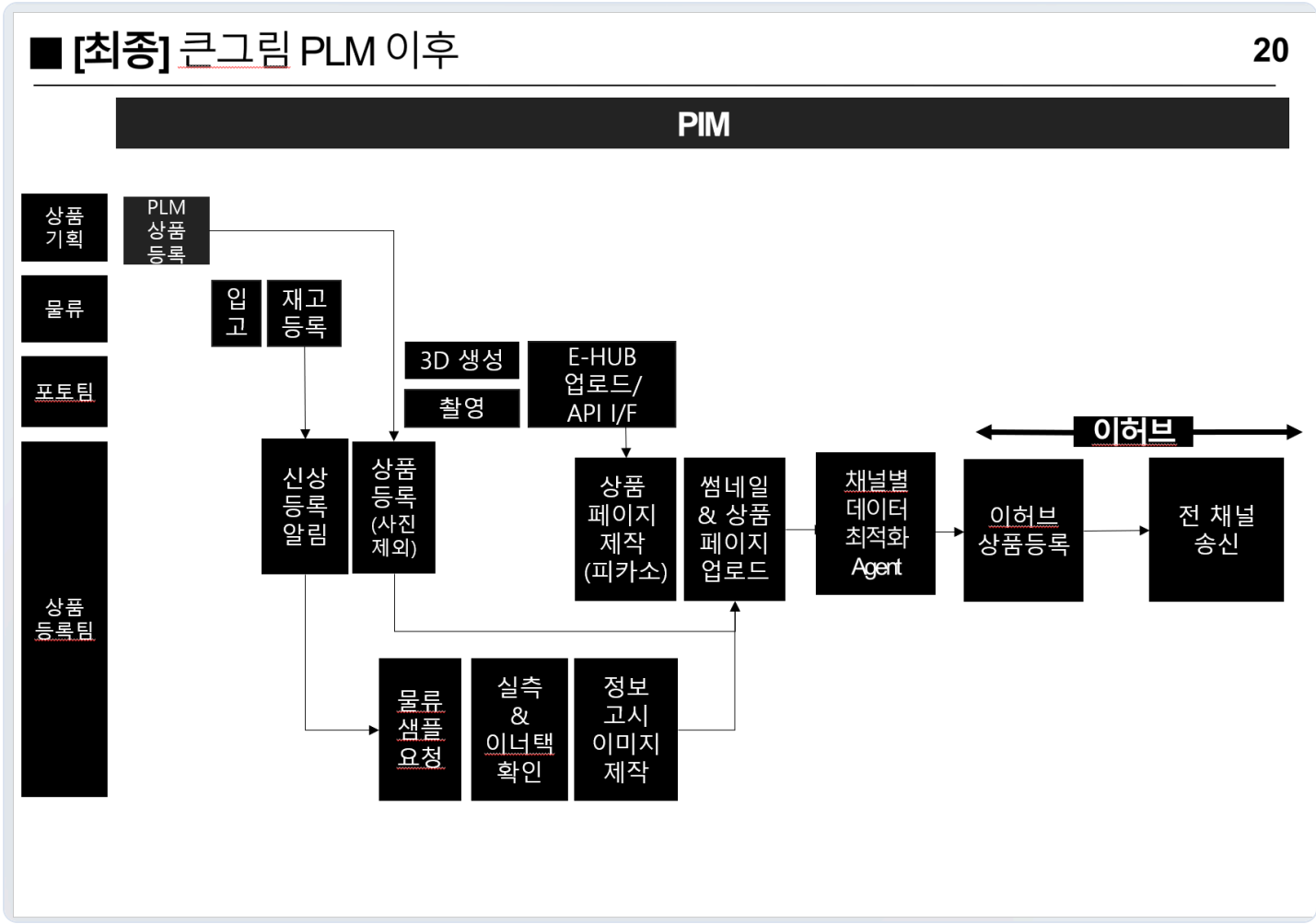
Integrated image platform pipeline: product data in → AI processing (keywords, channel listings) → OMS → online channels

Company-wide Integrated Image Platform



[AS-IS] The old listing process: the same information re-typed by people across five teams. The seven red boxes are the bug points

Company-wide Integrated Image Platform



[TO-BE] The image platform flow after PLM: new-product alerts go out automatically, shots and 3D renders enter through the hub, AI builds the pages, a per-channel agent fits each format and the listing hub sends to every channel. Every red box (bug point) is gone

Part 03 | Internal Business Systems Planning

07

Internal Strategy-Document RAG Search

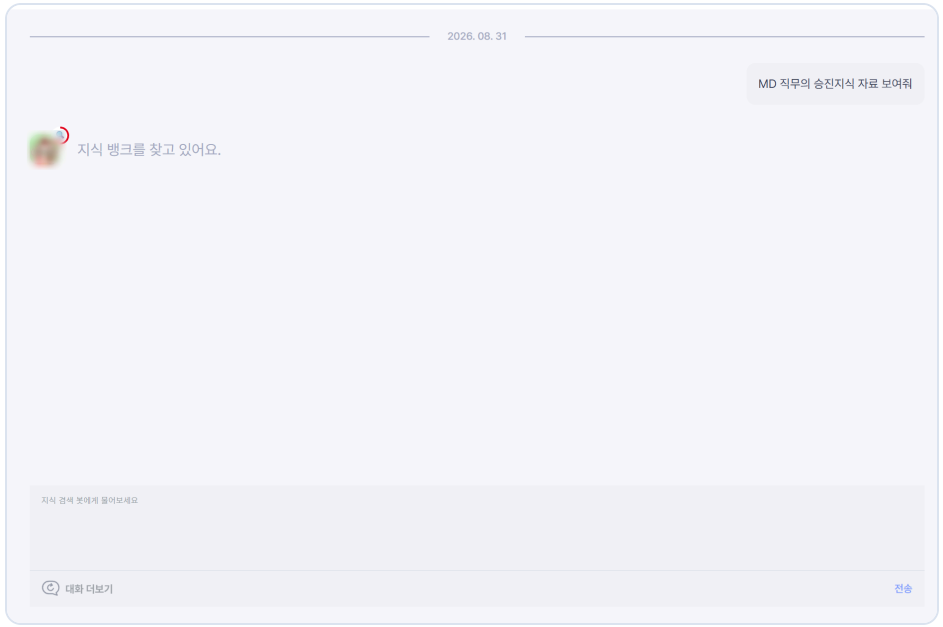
Not simple search, but a service that searches the organization's decision criteria

Period 2025.01 – 2025.07

RAG | Vector search | Knowledge management

Internal Strategy-Document RAG Search

Not simple search, but a service that searches the organization's decision criteria



Finding strategy documents in plain language

Period

2025.01 – 2025.07

Area

RAG · Vector search · Knowledge management

Stack & tools

OCR · Vector DB · GPT API

Purpose

Turned sales data, product information and internal project documents into an AI search and summary service.

Role & contribution

1. Converted strategy documents into JSON
2. Planned the UI/UX of the search screen
3. Iterated on preprocessing and retrieval logic through repeated user tests

Results

Budget saved	Search accuracy	Satisfaction	Usage in core teams
98%	**90%+**	**4.9 / 5**	**7× / week**
Self-built vector DB vs. allocated budget		N=68	N=68

Internal Strategy-Document RAG Search

1 Problem	2 What I did	3 Result
15 years of strategy decks were image-heavy PPTs that no one could search.	**Built natural-language search on OCR and a vector DB that answers with its sources.**	**90%+ search accuracy, 4.9/5 satisfaction and 98% of the allocated budget saved.**

Background

An asset without a way to use it: fifteen years of high-quality strategy material had accumulated, but image-heavy PPTs and file-level storage made it hard to search and use.

System architecture

Internal documents become JSON, vectorized with the OpenAI Embeddings API into a self-built vector DB. Only chunks close to the question go to the GPT API, which answers with sources.

① Internal source files

Fifteen years of strategy decks, many image-only and unsearchable as is.

- **PPT · Excel · PDF · Word**
- **OCR / image-to-text**
  text inside charts and images

② Document processing and JSON conversion

Split into search units and saved as JSON with sources.

- **Text extraction**
- **Document splitting**
  chunks per page or slide
- **Source metadata**
  file name · page · owning department · date

③ Building the search data

Vectorized with the OpenAI Embeddings API and stored in the self-built vector DB.

- **Embedding generation**
  OpenAI Embeddings API
- **Vector DB storage**
  self-hosted vector DB (98% below the allocated budget)

④ User question

Plain-language questions, e.g. “find last year's F/W sale strategy deck”.

- **Natural-language question**
- **User and department**
  used for permission filters

Internal Strategy-Document RAG Search

⑤ Question search

Question vector → similar-chunk search → top results, in in-house search code.

Search code (built in-house)
takes the question, runs the search, organizes results

Embedding model
vectorizes the question the same way as the documents

Vector search
similar chunks from the self-built vector DB, with source and relevance score

Result filtering and ranking
top results only, limited by department, date and permission

⑥ GPT API call

Only retrieved content goes to the model, told not to invent beyond it.

User question + retrieved document content + answer instructions

⑦ Final answer

Shown with source documents and pages for a jump to the original.

Answer

Source documents and pages

How a question is processed

1 Question
Ask in plain language.

2 Question embedding
Vectorized with the same model as the documents.

3 Similar-document search
Closest chunks retrieved from the vector DB.

4 Filtering and ranking
Top results kept, then filtered by permission and more.

5 GPT API call
Question + documents + instructions produce the answer.

6 Answer with sources
Source documents and pages shown below the answer.

Process

01 **Unstructured knowledge** › 02 **OCR / image-to-text** › 03 **Vector DB**

04 **GPT API natural-language search** › 05 **Iterative user testing**

Part 04 | Strategy Projects (Internal Consulting)

08

Redefining the Naver Channel Sales Strategy

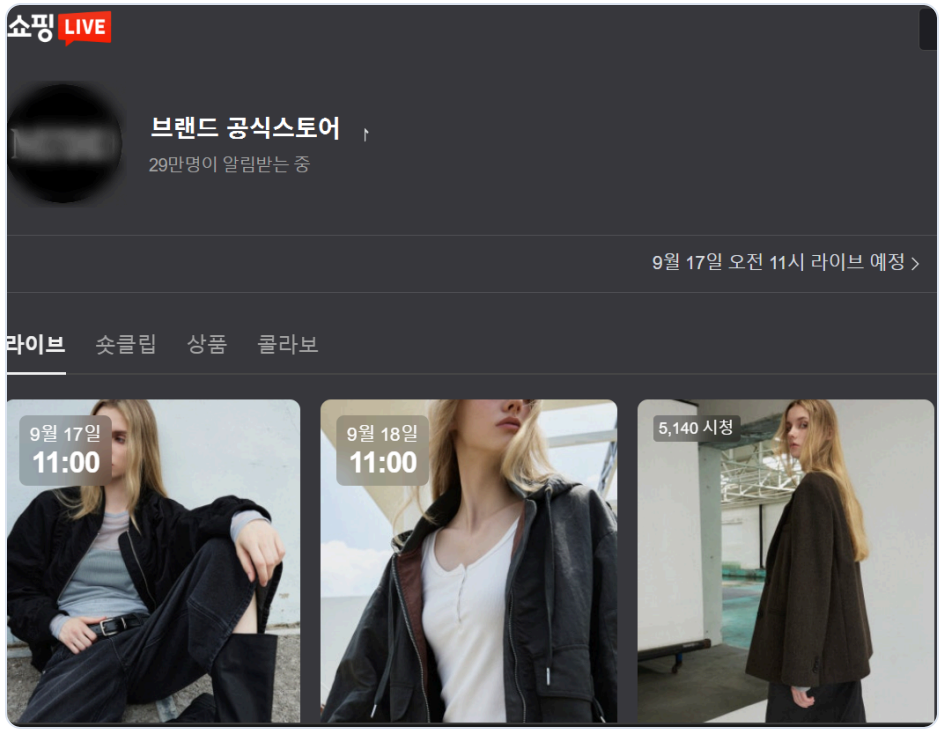
Why it was customer-structure redesign, not channel expansion

Period 2024.01 – 2024.02

Customer analysis | Channel strategy | Live commerce

Redefining the Naver Channel Sales Strategy

Why it was customer-structure redesign, not channel expansion



Naver Shopping Live channel

Period

2024.01 – 2024.02

Area

Customer analysis | Channel strategy | Live commerce

Purpose

Redesigned product, content and channel messaging from an analysis of each customer group's entry motives, lifting channel sales.

Role & contribution

1. Reinterpreted the limits of platform-led growth as a customer-structure problem
2. Analyzed purchase criteria and entry motives of customers in their 40s
3. Redesigned content messaging, review strategy and pricing around customers' purchase criteria

Results

Live-commerce conversion growth	Live-commerce sales per show
137%	**190%**
YoY, 0.8% → 1.9%	YoY, ₩100M → ₩290M

Redefining the Naver Channel Sales Strategy

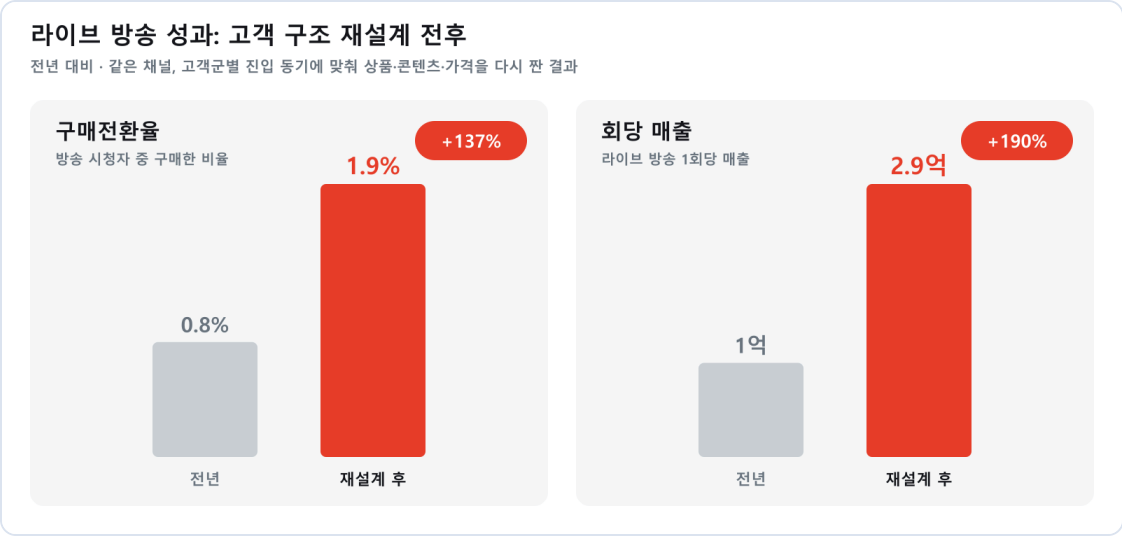
1 Problem	2 What I did	3 Result
Growing online sales called for a strategy to expand the brand into a new online channel.	**Analyzed why online customers in their 40s buy; redesigned products, content and pricing.**	**Live conversion rose from 0.8% to 1.9%; sales per show from ₩100M to ₩290M.**

Glossary

Conversion rate · Share of visitors who actually buy

Live commerce · Selling products through real-time video broadcasts

Entry motive · Why a customer came into the channel



Live commerce results: conversion 0.8% → 1.9%, sales per show 100M → 290M won (year on year)

Background

Growing online sales called for expanding into new online channels with a strategy to match. Online sales had been concentrated among customers aged 20–35, so a strategy to broaden both online channels and online customers was needed.

Part 04 | Strategy Projects (Internal Consulting)

09

Sports Brand Padded-Jacket Rebranding Strategy

Designing core products from big data

Period 2024.11 – 2024.12

Product strategy | Review analysis | Search trends

Sports Brand Padded-Jacket Rebranding Strategy

Designing core products from big data



Market analysis: shrinking padding demand, growing off-season padding demand

Period

2024.11 – 2024.12

Area

Product strategy | Review analysis | Search trends

Purpose

Redefined product strategy and marketing messages from sales, review and platform data by age group.

Role & contribution

1. Analyzed men's and women's needs for padding items
2. Analyzed three years of our own padding sales by item
3. Analyzed reviews of top-selling items on the three key platforms

Results

Sales growth among customers under 30

100%

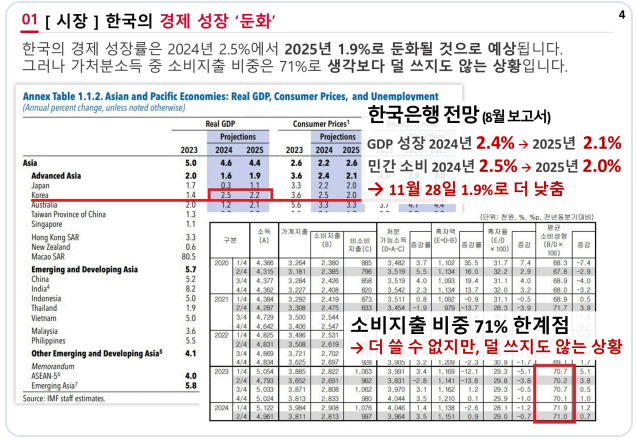
25FW vs. 24FW

Sports Brand Padded-Jacket Rebranding Strategy

1 Problem	2 What I did	3 Result
Planning repeated last year's best-sellers and missed why young buyers chose jackets.	**Analyzed 3 years of our own sales plus competitor and platform review data to redefine core products and messaging.**	**Sales among customers under 30 grew 100% (year on year).**

Glossary

Benchmark · A competitor case studied as a model to learn from
Synthetic fill · Padding made of polyester and other man-made fibers
Off-season padding · Padded jackets bought outside winter
Lightweight jacket · A thin, light padded jacket



[Market] Slowing growth: 1.9% forecast for 2025, consumer spending at a 71% ceiling of disposable income

Background

Customers wanted better value for the same money, and padded jackets were shifting from natural to synthetic fill.
Our planning carried over last year's top styles, missing customers under 30 and men in their 30s.

Four findings

[Market] Growth slows, spending at its ceiling	1.9% growth forecast for 2025, spending at 71% of income. Not closing wallets, but choosing better value for the same money.
[Market] Synthetic fill rising every year	In top-ranked jackets, natural fill fell 82% → 37% and synthetic rose 18% → 63% (2022 → 2024). That is why prices fell.
[Benchmark] Falling Nuptse, rising Onball	The North Face Nuptse -38%, Himalayan -37%, synthetic-fill Onball +57%. A value choice weighing warmth and price.
[Benchmark] Men in their 30s choose The North Face and Arc'teryx	Old benchmarks skew to 40s and 70%+ women, like ours. The North Face and Arc'teryx lightweight jackets: 46% and 67% men, led by 30s.

Sports Brand Padded-Jacket Rebranding Strategy

Process

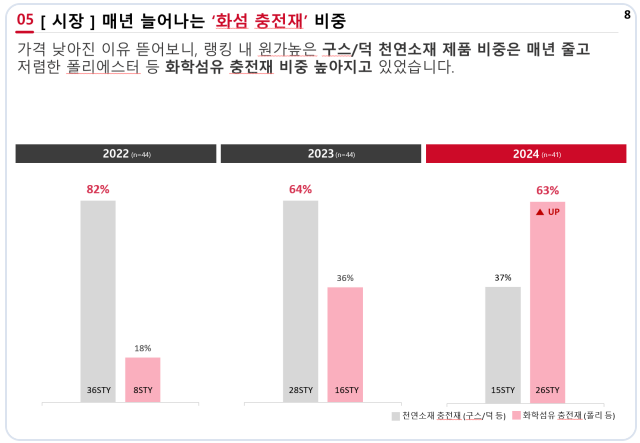
01 Market: economy and spending

02 Market: fill materials of top-ranked jackets

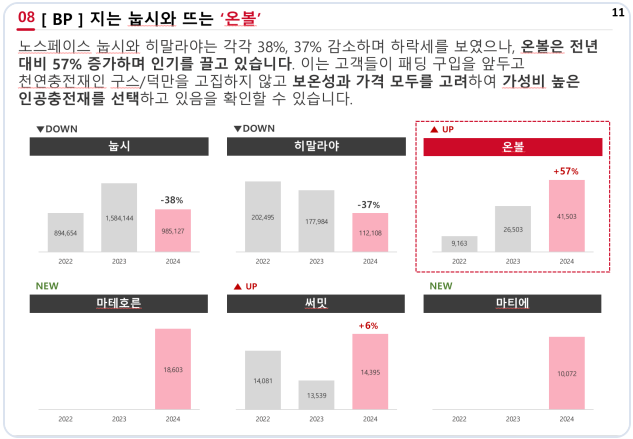
03 Benchmark: competitor product trends

04 Benchmark: age and gender preference

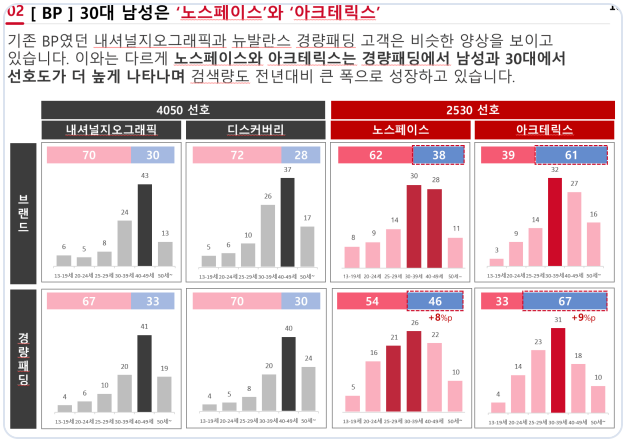
05 Redefine core products and message



[Market] Synthetic fill rising every year: natural (goose/duck) 82% → 64% → 37%, synthetic 18% → 36% → 63%



[Benchmark] Falling Nuptse, rising Onball: down jackets decline while the synthetic-fill Onball grows 57%, a value choice weighing warmth and price together



[Benchmark] Men in their 30s choose The North Face and Arc'teryx: the old benchmarks skew to 40s and women, the new ones to 25–39 and men

Part 04 | Strategy Projects (Internal Consulting)

10

Customer Churn Analysis for a Premium Brand in China

Six non-customer studies to find ‘brand casual’, localized for white-collar (白领) women

Period 2025.02 – 2025.04

Overseas market | Customer research | Product localization

Customer Churn Analysis for a Premium Brand in China

Six non-customer studies to find ‘brand casual’, localized for white-collar (白领) women



Store in China

Period

2025.02 – 2025.04

Area

Overseas market | Customer research | Product localization

Purpose

Cross-checked non-customer interviews, street-style surveys, review big data and social media to find why sales fell and whom to win, carried through to SS and FW product input.

Role & contribution

Proposed cutting formal-led products and rebuilding around shirts, denim and business casual.

1. Defined the cause of declining sales as a shift in customers' workwear norms, not the products
2. Ran six customer-research methods from zero, revising questions and methods each round, and wrote them up as the product customer-research validation process
3. Derived steady and trend types per item and customers' top needs and complaints from three years of Tmall top-20 sales data and review big data
4. Documented basics and differentiators for dresses, denim, knits/cardigans, shirts, tees and jackets by learning from benchmarks (Ralph Lauren, Edition, MO&Co., Uniqlo)
5. Decided SS and FW inputs (type, fit, color, fabric) with the local merchandising team

Results

Sales growth vs. same month last year

+8%

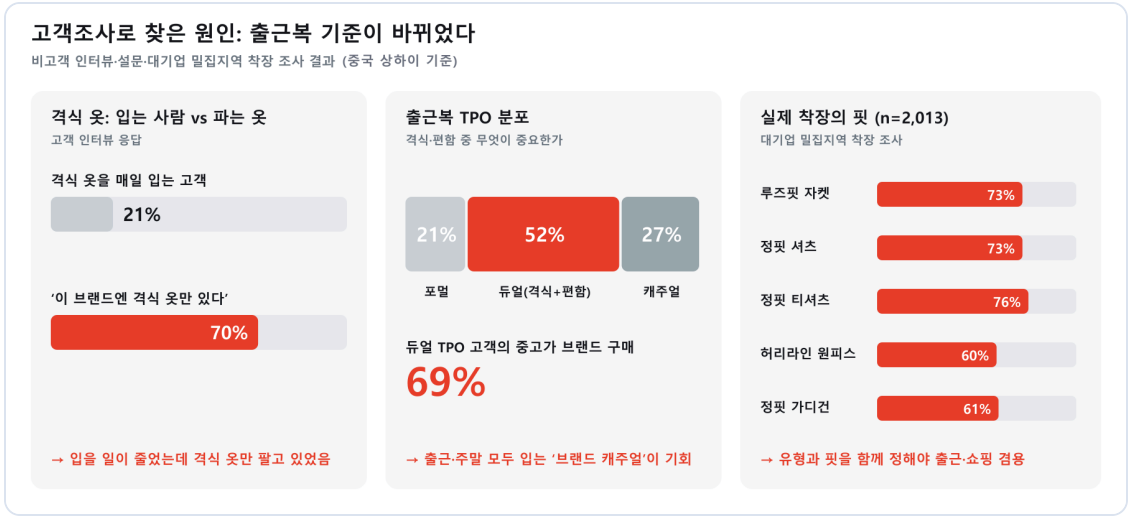
from decline to growth

Customer Churn Analysis for a Premium Brand in China

1 Problem	2 What I did	3 Result
Brand sales fell 27% for two years in a row.	**Ran six customer studies, including non-customer interviews, and localized the products.**	**Sales turned to growth, up 8% on the same month last year.**

Glossary

Non-customer research · Studying people who do not buy the brand
White-collar (白领) · Chinese term for office workers
TPO · Time, place and occasion an outfit is worn for
Brand casual · Premium casual wear fit for both office and weekend
SS · FW · Spring/summer and fall/winter seasons
Shop-along · Shopping with a customer to observe how they buy



Key research numbers: formal demand 21% vs 70% supply perception, dual TPO 52%, actual outfit fits

Background

The brand kept decade-old hits such as glossy colored blouses, and sales fell 27% for two years in a row.
Office dress had gone casual, but only formal suiting remained: clothes 'worn one day a week'.

Findings and strategy

[Research] Six customer studies + three-year trends	Cross-checked interviews, shop-alongs, surveys, dry-cleaner counts, outfits and review big data. Three-year sales trends, not last season's bestsellers.

Customer Churn Analysis for a Premium Brand in China

[Diagnosis ①] Wearable one day a week, so a waste of money	Office dress went casual; the brand stayed formal. 21% wore formal clothes daily, 70% said it had only formal clothes.
[Diagnosis ②] The customer to win: global-minded white-collar women	Women aged 35–45 at large companies, 44% global-brand buyers. The brand's gold trims and loud colors read as 'old'.
[PART ① brand casual] Expand to the dual TPO of daily commute and weekend shopping	Dual TPO, where formality and comfort both matter, is largest at 52%, with 69% mid-to-high-priced brand purchases: the opportunity.
[Type and fit] Relaxed fit for formal types, regular fit for the rest	Regular fit reads formal, looser reads casual. Of 2,013 outfits: loose-fit jackets 73%, regular-fit shirts 73%.
[PART ② Localization] At this price, versatility and quality-led luxury	Good design is the baseline; versatility and quality-led luxury decide the purchase. Positioned between SPA and premium brands.
[Execution] SS and FW inputs	SS: brand casual types added (delivered 31 March). FW: 4 denim styles (order value up 320%), 15 shirt styles, 6 shirt-dress styles.

Process

01 **Six non-customer studies** › 02 **Diagnosis: once-a-week wear · customer mismatch** › 03 **PART ① brand casual principles**

04 **PART ② Localizing to white-collar needs** › 05 **Basics and differentiators per item** › 06 **SS · FW product input**